Bharshita Deka (230123016)
Mark: 11/15

+10/1/42+

MA 324 | Quiz III
Statistical Inference and Multivariate Analysis | 16:05–16:50 IST
IIT Guwahati | April 08, 2026

Answers must be given ***exclusively on this sheet:*** *answers given on the other sheets* ***will be ignored****. This question paper has* ***6 questions*** *and 2 pages.*

Name and Roll Number

Please mark your Roll No:

○0 ○0 ●0 ○0 ○0 ○0 ●0 ○0 ○0
○1 ○1 ○1 ●1 ○1 ○1 ○1 ●1 ○1
●2 ○2 ○2 ○2 ●2 ○2 ○2 ○2 ○2
○3 ●3 ○3 ○3 ○3 ●3 ○3 ○3 ○3
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○5 ○5 ○5 ○5 ○5 ○5 ○5 ○5 ○5
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○7 ○7 ○7 ○7 ○7 ○7 ○7 ○7 ○7
○8 ○8 ○8 ○8 ○8 ○8 ○8 ○8 ○8
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Name of the Student: Bharshita Deka

Roll Number of the Student: 230123016

Signature of the Student: Bharshita Deka

Signature of the Invigilator:

Questions and Responses

Question 1: (3 points) For $Y \in \mathbb{R}^n, X \in \mathbb{R}^{n\times p}$, and $\beta \in \mathbb{R}^p$, consider a regression model

$$Y = X\beta + \epsilon,$$

where ϵ has an n-dimensional multivariate normal distribution with zero mean vector and identity covariance matrix. Let I_p denote the identity matrix of order p. For $\lambda > 0$, let

$$\widehat{\beta}_n = (X^TX + \lambda I_p)^{-1}X^TY,$$

be an estimator of β. Then which of the following options is/are correct?

- ○ $\widehat{\beta}_n$ is an unbiased estimator of β
- ○ $\text{Var}(\widehat{\beta}_n) = (X^TX + \lambda I_p)^{-1}$
- ☒ $\widehat{\beta}_n$ has a multivariate normal distribution
- ☒ $(X^TX + \lambda I_p)$ is a positive definite matrix

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Question 2: (3 points) Consider the multiple linear regression problem

$$Y = X\beta + \epsilon, \text{ and } \hat{Y} = X\hat{\beta}$$

where $Y = (Y_1, \cdots, Y_n)'$ is a $n \times 1$ vector of response variable, X is a $n \times p$ design matrix with full rank (largest possible rank), β is a $p \times 1$ vector of parameters, and ϵ is a $n \times 1$ vector of errors; $n > p$. The first element of β is the intercept of the model. Then which of the following statements is/are true?

- ☒ $\sum_{i=1}^n Var(\hat{Y}_i) < \sigma^2 n$ ✓
- ○ $\sum_{i=1}^n Var(\hat{Y}_i) = \sigma^2 n$
- ☒ $\sum_{i=1}^n Var(\hat{Y}_i) = \sigma^2 p$ ✓
- ○ $\sum_{i=1}^n Var(\hat{Y}_i) < \sigma^2 p$ ✗

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Question 3: (2 points) Let a linear model $Y = \beta_0 + \beta_1 X + \epsilon$ be fitted to the following data, where ϵ is normally distributed with mean 0 and unknown variance $\sigma^2 > 0$.

x_i	0	1	2	3	4
y_i	3	4	5	6	7

Let $\widehat{Y}_0$ denote the ordinary least-square estimator of Y at $X = 6$. If $E\left(\widehat{Y}\right) = c_1\beta_0 + c_2\beta_1$ and $Var\left(\widehat{Y}_0\right) = d\sigma^2$, then which of the following is/are true?

0/2

- $c_1 = 6$, $c_2 = 1$, $1.7 \le d \le 1.9$
- $c_1 = 1$, $c_2 = 6$, $1.7 \le d \le 1.9$ ✓
- $3.1 \le c_2 - c_1 - d \le 3.5$ ✓
- $c_1 + c_2 + d \le 8.8$ ✓

Question 4: (3 points) Let $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \cdots + \beta_{22} x_{22,i} + \epsilon_i, i = 1, 2, \cdots, 123$, where, $x_{1,i}, \cdots, x_{22,i}$ are fixed input (independent) variables; and, for $j = 0, 1, 2, \cdots, 22$; β_j are unknown parameters, and ϵ_i's are independent and identically distributed normal random variables with mean zero and finite variance. If $SS_{Reg} = 338.92$ and $SS_T = 522.30$ and the value of R^2_{adj} (adjusted R^2) equals to t. Then which of the following statements is/are true?

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- $0.51 < t < 0.59$ ✓
- $0.53 \le t < 0.63$ ✓
- $0.61 < t < 0.69$
- $0.43 < t < 0.54$

Question 5: (2 points) Consider a multiple linear regression problem. Which of the following statements is/are true?

0/2

- A linear model is linear in the parameters, but not necessarily in the independent variables ✓
- Q-Q plot is used to test for normality ?
- A leverage point is always influential point ✗
- PRESS is a measure of how well a regression model perform in predicting new observations ✓

Question 6: (2 points) Consider the linear model

$$
\begin{aligned}
Y_1 &= \beta_1 + \beta_2 + \epsilon_1 \\
Y_2 &= \beta_1 + 2\beta_2 + \epsilon_2 \\
Y_3 &= \beta_1 + c\beta_2 + \epsilon_3,
\end{aligned}
$$

where $\beta_1, \beta_2 \in \mathbb{R}$ are unknown parameters, and the uncorrelated errors $\epsilon_i, i = 1, 2, 3$ have zero mean and finite variance $\sigma^2 (> 0)$. The constant c is such that $\hat{\beta}_1$ and $\hat{\beta}_2$ are uncorrelated, where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the best linear unbiased estimators of β_1 and β_2, respectively. Then $(\mathrm{Var}(\hat{\beta}_1), \mathrm{Var}(\hat{\beta}_2))$ equals

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- $(\frac{\sigma^2}{3}, 14\sigma^2)$
- $\left(\frac{\sigma^2}{14}, \frac{\sigma^2}{3}\right)$
- $(3\sigma^2, \frac{\sigma^2}{14})$
- $\left(\frac{\sigma^2}{3}, \frac{\sigma^2}{14}\right)$